PET BINARY CLASSIFICATION :

Numerical Metrics

Split	ROC-AUC	Accuracy	Balanced Accuracy	Sensitivity / Recall	Specificity	Precision	F1-score	Brier Score	PR-AUC	TN	FP	FN	TP	Expected Calibration Error / ECE
validation	0.4864	0.7662	0.4828	0.8406	0.1250	0.8923	0.8657	0.2338	0.8929	1	7	11	58	0.2338
test	0.5395	0.7821	0.4903	0.8696	0.1111	0.8824	0.8759	0.2179	0.8941	1	8	9	60	0.2179

- ROC-AUC is around 0.52 to 0.54, which is near random. This means the PET probability ranking is weak.

- Accuracy is lower than the clinical model because PET predicts more negative cases and therefore makes more mistakes on the dominant positive class.

- Balanced accuracy is slightly better than clinical because PET has nonzero specificity, but it is still weak.

- Sensitivity is acceptable but not ideal.

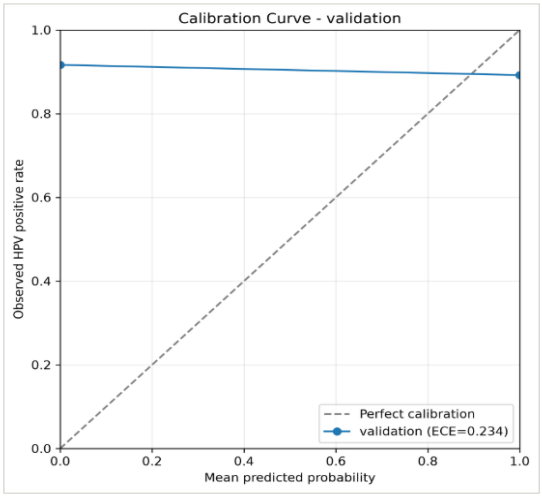
- Specificity remains poor, so the model is not dependable for identifying HPV-negative patients.

- PR-AUC is near 0.89 to 0.90, but the positive class baseline is already about 0.89, so this is only a small improvement.

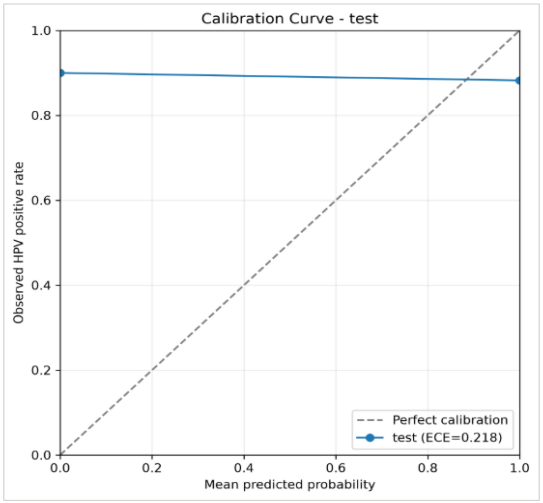
- Brier score and ECE are worse than clinical. This means PET probabilities are poorly calibrated.]

-ROC-AUC is only slightly above random on validation and test. Balanced accuracy is also only slightly above 0.5.

Calibration Curve

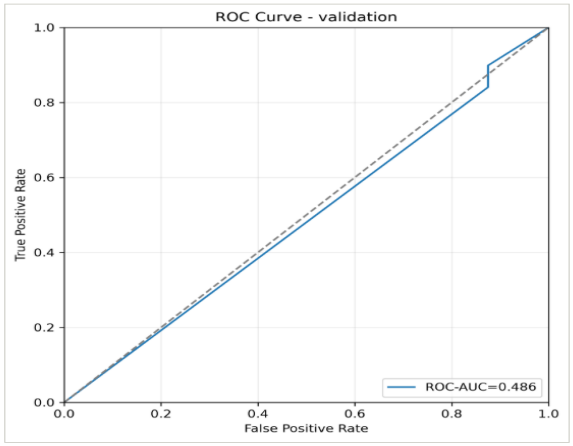


Calibration Curve

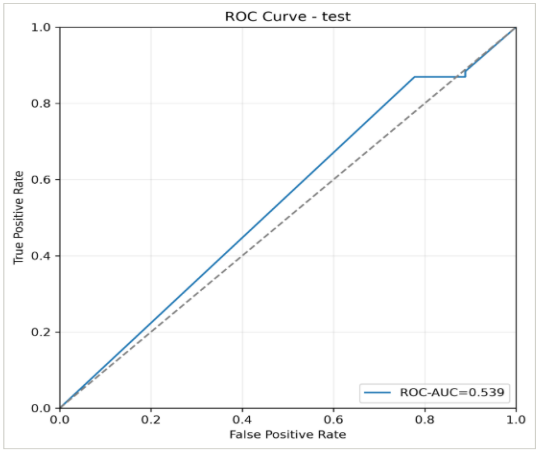


The PET calibration table shows extreme predicted probabilities. Many patients fall into near 0.0 or near 1.0 probability bins. In both validation and test, the low-probability bin still contains mostly HPV-positive patients. For example, in validation, the [0.0, 0.1) bin has an observed HPV-positive rate of 0.857. This means the model is confidently wrong for many cases. That explains the high Brier score and ECE.

ROC Curve

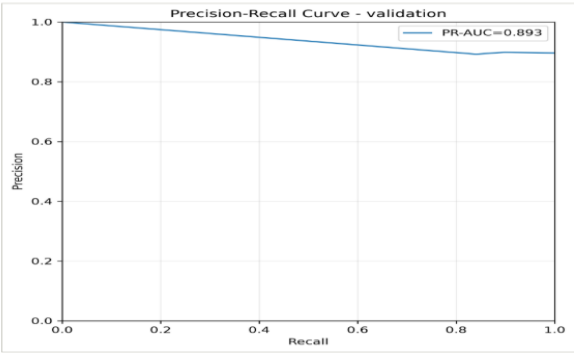


ROC Curve

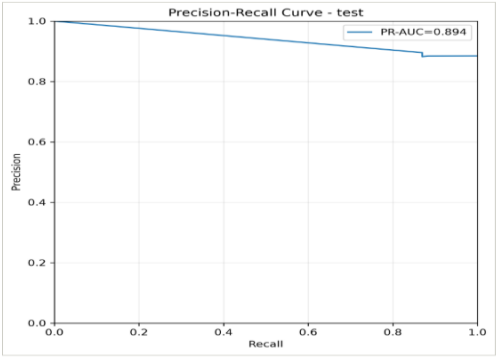


The PET ROC curve should ideally rise well above the diagonal random line. Instead, validation ROC-AUC is 0.535 and test ROC-AUC is 0.519, so the ROC curves are only slightly better than random. This means the PET model has very weak ranking ability. It cannot reliably assign higher probabilities to HPV-positive patients than to HPV-negative patients.

Precision-Recall Curve

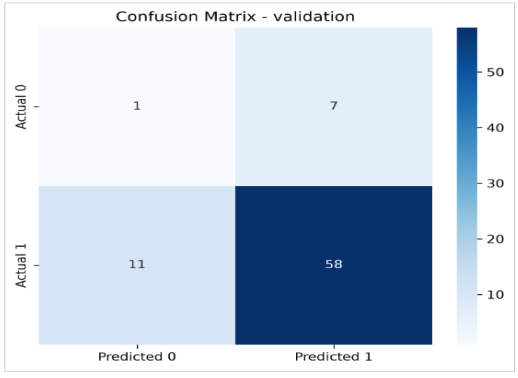


Precision-Recall Curve

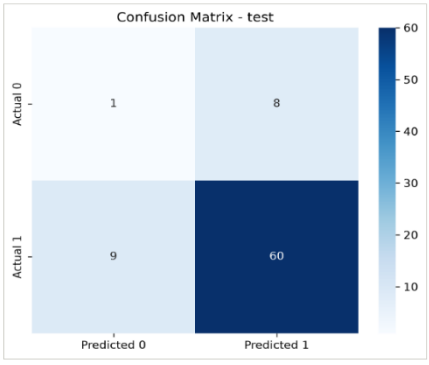


The PR-AUC values are 0.904 on validation and 0.890 on test. These look high, but the dataset is already about 89 percent HPV-positive. That means the PR curve is close to the natural positive-class baseline. The PR plot should therefore not be read as strong evidence of good PET classification.

Confusion Matrix Heatmap

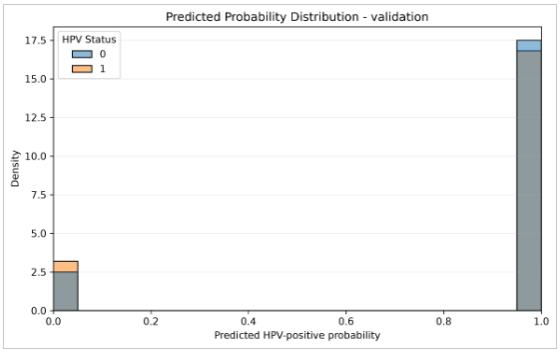


Confusion Matrix Heatmap

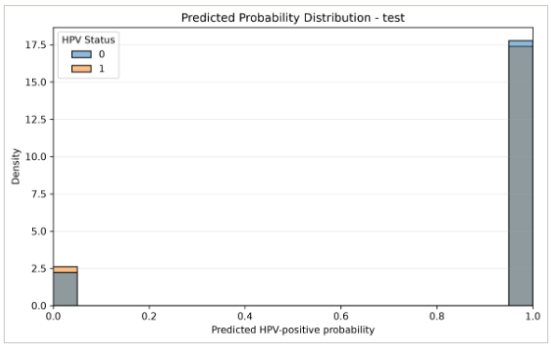


The PET confusion heatmap is better than the clinical heatmap because it has 2 true negatives on validation and 2 true negatives on test. However, it still has many errors: 6 false positives and 12 false negatives on validation, and 7 false positives and 12 false negatives on test. The heatmap shows that the model is unstable: it detects some negatives, but itlso sacrifices many positive caseS.

Probability Distribution

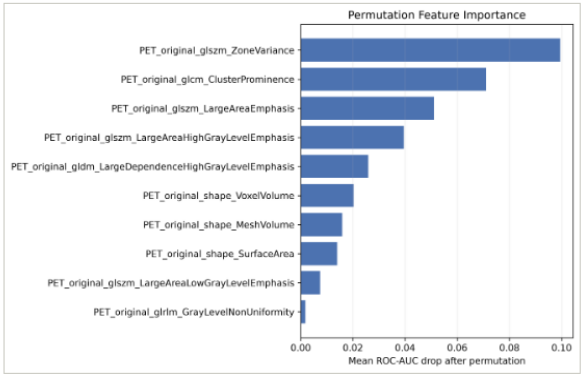


Probability Distribution

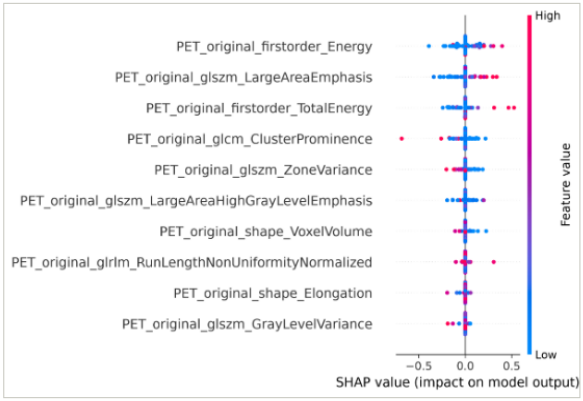


An ideal PET probability plot would show a clean separation between HPV-negative and HPV-positive probability distributions. The PET plot instead shows extreme probabilities near 0 and 1. This is not good separation; it is overconfidence. The low-probability region still contains many HPV-positive cases, which is why calibration and Brier score are poor.

Feature Importance Plot



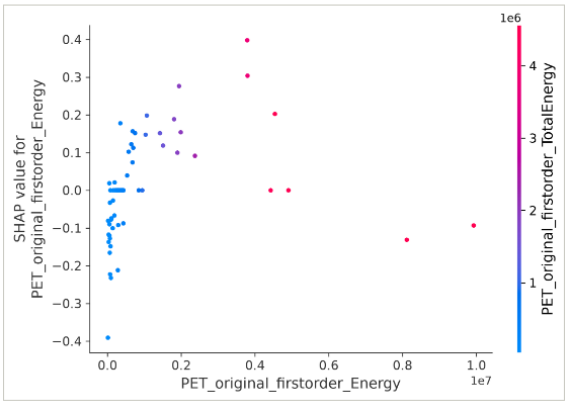
SHAP Summary Plot



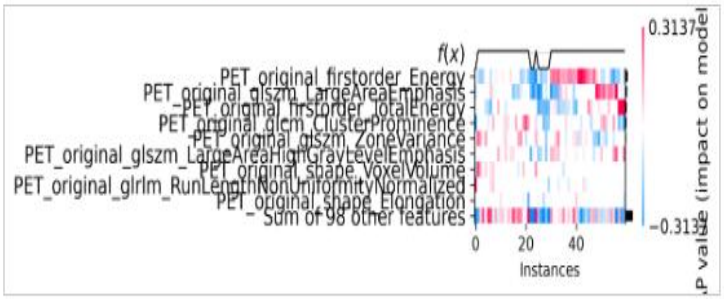
The permutation importance plot is dominated by PET energy, volume, and texture features, especially TotalEnergy, MeshVolume, LargeAreaHighGrayLevelEmphasis, and SurfaceArea. This tells us the model uses PET radiomics signal, but the weak ROC-AUC means those features do not produce robust HPV discrimination.

The SHAP summary plot confirms that TotalEnergy and Energy dominate the model explanations. These features may reflect tumor intensity or tumor burden, but they do not reliably separate HPV status in this dataset. A few large PET features drive predictions, which can make the model fragile.

SHAP Dependence Plot



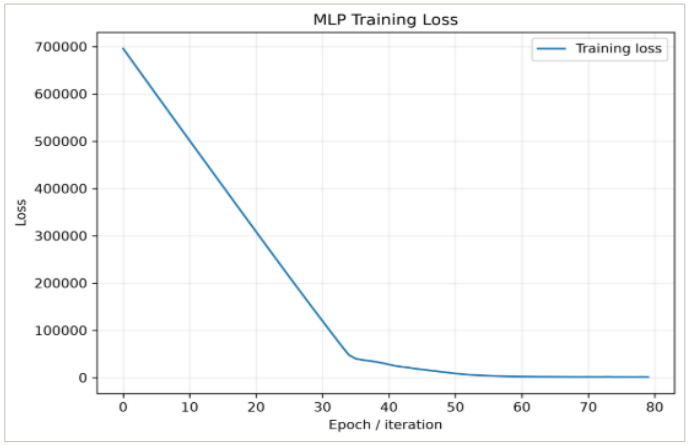
SHAP Heatmap Plot



The dependence plot automatically selected PET_original_firstorder_TotalEnergy. This indicates TotalEnergy has the strongest average SHAP effect. However, a strong SHAP effect does not automatically mean good classification. Here, the model strongly uses TotalEnergy, but the final ROC-AUC remains near random.

The PET SHAP heatmap shows that the model's explanations are concentrated in a small group of radiomics features. Ideally, we would want consistent SHAP patterns that separate correctly classified positive and negative cases. Here, the heatmap supports the idea that the model is relying on a few high-scale radiomics variables rather than a stable multi-feature pattern.

Training Loss Curve



The training loss curve can show that the neural network trained, but it does not prove good generalization. PET radiomics features have very large numerical ranges and were not scaled, following the project rules. This can make MLP training harder and can contribute to unstable, overconfident probabilities.